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First implementation of a Gadget model for the analysis of hake in the Mediterranean

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ABSTRACT

An age-length structured model was built for European hake *Merluccius merluccius* in the central Mediterranean Sea using Gadget. This analytical framework allowed to integrate multiple sources of information, including fisheries-dependent and fisheries-independent data, collected at different scales and aggregation levels. The model includes the two main fisheries targeting hake in the area, the trawl and the gillnet fisheries, and account for differences in their selectivity and effort. Alternative models were used for testing different assumptions on recruitment and growth. The model is then used to predict the main trajectory of the stock during the next years and to evaluate the potential effects of implementing closed areas management scenarios in the hake nurseries as a specific tool to reduce fishing mortality on recruits. The modelling framework presented performed successfully also in a commercial landing data limited context, common for the Mediterranean. Our results provide statistical support for fast growth and multiple recruitment events assumptions. Including both these key features represent an unprecedented improvement of modelling hake population dynamics in the Mediterranean. We found that the reduction in the fishing effort that characterized the fisheries in the study area during the last few years, coupled with fast recovery abilities of the hake stock, has the potentiality to allow a moderate increase of the stock during the next years. Interestingly, our simulations show that the positive effects which might be expected from protecting hake nursery grounds are only marginally related to a reduction in hake recruits fishing mortality. Although our model relies on assumptions and surely represents an over-simplification of the real world, it still contributed to improve our understanding of the temporal dynamics of one of the most valuable fish stocks in the central Mediterranean.

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1. Introduction

One of the main objectives of statistical modelling is to integrate multiple sources of information and use them simultaneously to estimate parameters. The advantages of using information from different datasets within the same single model fitting procedure have been clearly expressed by several authors (i.e. Demyanov et al., 2006; Methot, 1989) and widely recognised, but various problems are related to this issue (Richards, 1991; Stefansson, 2003).

The study of dynamics of fish populations requires incorporating a variety of processes and multiple datasets. Many widely used assessment models (e.g. XSA – Shepherd, 1999) do not consider this aspect and almost ignore combination of multiple datasets and/or combine these non-statistically prior to fitting a model. An appropriate integration of different components of a model (e.g. stocks, fleets), generally described by different variables from different datasets, can be done through the use of (weighted) likelihood components (Taylor et al., 2007). This also provides a basis for implementing complex and disaggregated models (Taylor and Stefansson, 2004).

An age-length structured model was built for hake in the central and northern Tyrrhenian (central Mediterranean Sea), using Gadget (Begley and Howell, 2004; Begley, 2006). Gadget is the Globally applicable Area Disaggregated General Ecosystem Toolbox developed to model marine ecosystems, and it is able to integrate species interaction and the impact of fisheries exploiting the stocks of that species (Taylor et al., 2007). As specified by Stefansson and Palsson (1998), Gadget has been developed as a forward simulation model using statistical estimation through weighted combinations of several log-likelihood criteria. Among the main advantages of Gadget respect other widely applied stock assessment methods there are

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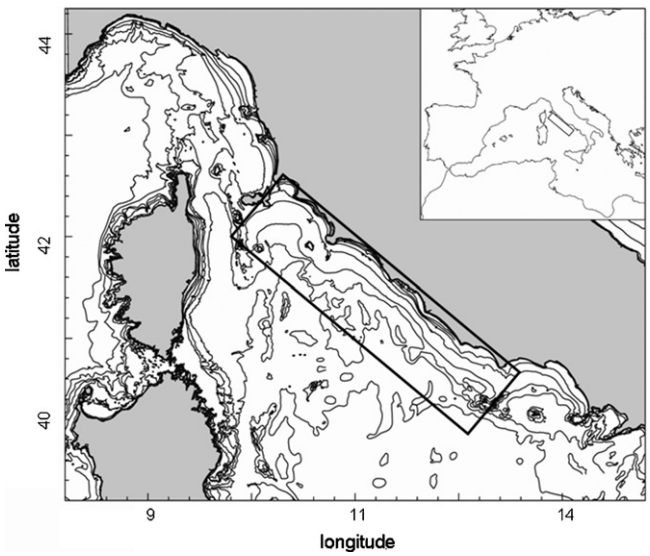


Fig. 1. Map of the study area.

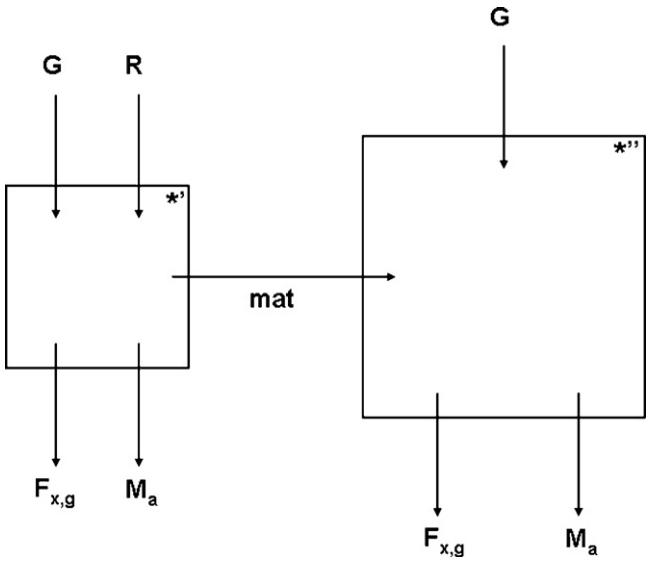


Fig. 2. Basic conceptual hake model simulating small (*′) and large (*″) hake stock components and the processes of growth (G), recruitment (R), natural mortality by age (M_a), fishing mortality by fish length and fishing gear ($F_{x,g}$), movement of fish from the small hake to the large hake component (mat).

its flexibility, low data-demanding, and the possibility to integrate into the same model incomplete time series of data at different aggregation levels and scales. These features make Gadget an ideal tool for modelling fish population dynamics in context with sparse and moderate amount of data such as in the Mediterranean.

The complex ecology of European hake, *Merluccius merluccius* (Alheit and Pitcher, 1995), the key role it plays within Mediterranean demersal assemblages (Colloca et al., 2004) and its economic importance (SGMED, 2008) attracted the attention of marine ecologists and fisheries scientists since long time. This produced an extensive number of studies and literature on European hake during the last decades that made it probably one of the most studied species in the Mediterranean basin. In spite of the available knowledge, very limited synthesis has been done in the form of models of population structure (Apostolaki et al., 2002).

Hake represents one of the most important commercial species targeted by demersal fisheries in the western Mediterranean. In the Tyrrhenian Sea (Fig. 1) several different gears target hake, including trawls, and gillnets with different mesh sizes. Several studies have demonstrated how trawls and gillnets in the Mediterranean have different selectivities and thus exploit different portions of hake stock (e.g. Aldebert et al., 1993; Abella and Serena, 1998; Colloca et al., 2000). The common use in the Mediterranean of trawl nets with 40 mm and smaller cod-end mesh size, which finds an explanation in the high market value of small hake and other incidentally caught species, results in trawl catches being dominated by small-sized hake, mostly immature. Comparative fishing experiments with different mesh sizes (mentioned in Caddy, 1990) suggested that larger hake can be potentially less vulnerable to fine mesh gears or they can distributed over areas where they are less vulnerable to trawling activity. Studies of other fishing methods in the Gulf of Lions (i.e. Aldebert et al., 1993) showed that the older part of the hake stock, composed of mature fish and representing the spawning potential, is poorly caught by trawlers on the continental shelf. This limited pool of mature fish is however vulnerable to gillnets and longlines, especially along the edge between shelf and slope.

In the Tyrrhenian Sea both trawlers and a small-scale gillnet fishery target hake (Abella et al., 1997). Trawlers exert a higher fishing effort than gillnets and they catch a larger number of hake in the Tyrrhenian Sea as well as in most of the north-western Mediterranean but, due to marked differences in the spectra of lengths targeted by the two gears, the biomass landed is quite comparable between these two fishing segments (official national statistics by IREPA – Institute of Economic Researches on Fishery and Aquaculture).

Estimation and monitoring of adult fish biomass is considered one of the most important aspect of fish stock assessment and management (Francis, 1997), since spawning fish contain the potential of a stock to renew itself and thus persist in time (SSB paradigm), large fish often occupy higher levels of the trophic web, and because large fish tend to be of greater economic value than small fish.

The main objective of this work is to develop a basic population structure model for hake in the Tyrrhenian Sea, testing on the way alternative model assumptions concerning recruitment, growth and fishing effort. The basic model is subsequently used to evaluate the potential effects of implementing closed areas management scenarios in the hake nurseries as a specific tool to reduce fishing mortality on recruits.

2. Materials and methods

A series of equations containing several unknown parameters define the model (Figs. 2 and 3). Gadget runs a forward prediction model using these equations and parameters, and then compares the outputs from the model to the observed data to get likelihood scores as a goodness of fit of each model component. Through a combination of different minimisation algorithms (Simulated Annealing, Hooke & Jeeves, BFGS) the parameters are adjusted until a minimum in the overall likelihood score is found (best fit of the model to the data). A fundamental part of the parameter estimation procedure consists of assigning a weight to each source of information (log-likelihood component) to define its contribution to the overall sum log-likelihood. The weighting method proposed by Stefansson (1998, 2003) is used in the following, where the importance assigned to each source of information or component is proportional to how well the model fits the data and to the size of the component itself.

The basic conceptual model of hake (Fig. 2) consists of two stock components represented by small (2–40 cm in TL, age 0–3) and large (40–100 cm in TL, age 1–4+) hake in a single area (central and northern Tyrrhenian). Internally, the model uses a quarterly time step for a total time span of 14 years (from 1994 to 2007).

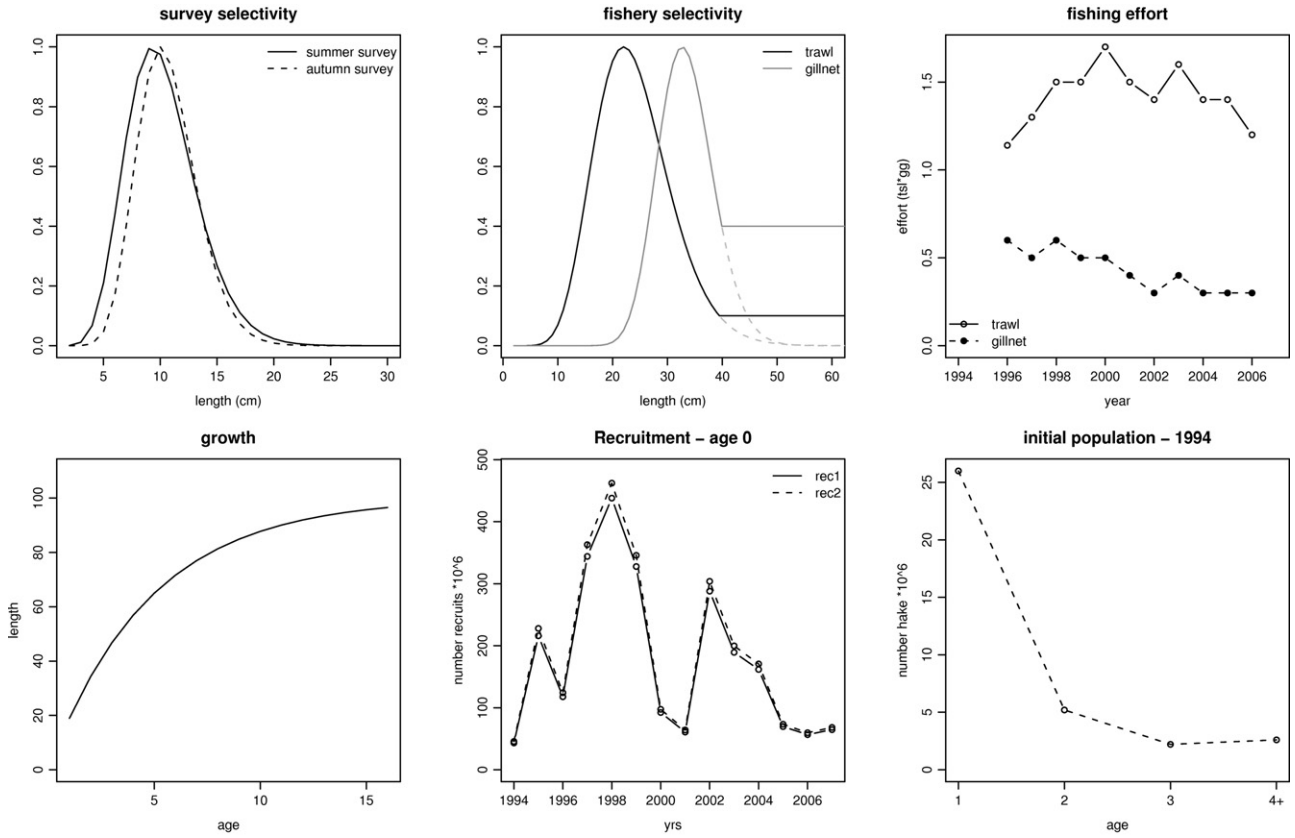


Fig. 3. Main model relationships (from top left) survey selectivity pattern, commercial trawler and gillnet selectivity pattern, trawler and gillnet fishing effort, von Bertalanffy growth function, number of recruits coming into the model, age structure of the initial population (timestep 1, year 1994).

The growth process (G) was modelled on the base of a simplified version of the von Bertalanffy growth equation as follow:

$$\Delta L_i = (L_\infty - L_i)(1 - e^{-k\Delta t})$$

$$\Delta W_i = q_1((L_i + \Delta_i)^{q_2} - L_i^{q_2})$$

where L is fish length, t is time and W is fish weight. The growth parameters ($L_\infty = 100$ cm; $k = 0.21$; $q_1 = 0.59 \times 10^5$; $q_2 = 3.0595$) were fixed according to the most recent assessments (SGMED, 2008) and this in agreement with the fast growth hypothesis proposed for hake (e.g. Garcia-Rodriguez and Esteban, 2002) on the base of recent results from tagging experiments (Pinero et al., 2004; De Pontual et al., 2006), reinterpretation of otolith rings (Belcari et al., 2006; Kacher and Amara, 2005; Morales-Nin and Moranta, 2004) and bioenergetic considerations (Jobling, 1994; Because, 2007).

Classical assessment for this stock has often assumed natural mortality to be constant. This could be an appropriate assumption, although a simplification, considering for instance that many North Atlantic fish stocks are mostly exploited when fish are mature and that small fish are generally moderately selected by fishing gears or that they have no economic value. The lack of validity of these assumptions for most of the Mediterranean demersal fisheries, including hake fisheries, has been pointed out by several authors (i.e. Caddy, 1993; Caddy & Abella, 1999; Colloca et al., 2000). In particular it is unreliable to assume a constant natural mortality rate for all age classes, mainly because younger individuals are susceptible to higher predation mortality and more intense density-dependent mechanisms. A rapid decline in natural mortality rate M should be expected in the first years of life towards a relatively more constant natural mortality for adult fishes. A natural mortality vector (M_a) was thus adopted in accordance with recent assessment of the hake stock in the area (SGMED, 2008: $M_0 = 1.3$; $M_1 = 0.8$; $M_2 = 0.4$; $M_3 = 0.3$; $M_{4+} = 0.2$).

Movement of fish (mat) from the small to the large hake component was modelled to occur at the fixed length of 40 cm. Given that the length-at-first maturity of female is 35 cm in TL (SGMED, 2008) the large hake component refers approximately to the mature portion of the stock. Two recruitments (R) scenarios are simulated in the first and second time step every year. They have the same pattern, based on standardised geostatistical estimations (Colloca et al., 2009), but different levels (using a scale factor estimated within Gadget, Fig. 3). Two fishing fleets are simulated in the model, the trawlers and the gillnet fisheries. The reduction of animals by the simulated fisheries is given by the combination of fishing level (effort) and a "suitability function" (gear selection), disaggregated by gear. A gear-specific time series of fishing effort (Fig. 3) was obtained from official statistics (IREPA, http://www.irepa.org/irepa/sistan/sisdati.html), based on the combination of two main components, fishing capacity (G^W: gross weight of vessels measured in tons) and fishing activity (D: mean number of fishing days by boat):

$$E = \frac{G^W \cdot D}{10^6}$$

Trawlers and gillnets displayed also a different pattern of selectivity (Fig. 3), targeting different portions of the stock, that was approximated through gear-specific parameters. The suitability function $S(L)$ for both the fishing gears was a combination of a Gamma and a constant function:

$$S(L) = \left(\frac{L}{(\alpha - 1)^{\beta\gamma}}\right)^{\alpha - 1} e^{(\alpha - 1 - L)/\beta\gamma}$$

$$S(L) = \delta$$

where L is fish total length and α, β, γ, and δ are the parameters of these functions, to be estimated (or set) for each gear.

2.1. Data and likelihood functions

For each observed dataset a corresponding fitted dataset can be generated by Gadget, allowing a numerical comparison between the modelled and observed populations. This gives the possibility to properly evaluate the fit of each model component to the available datasets.

The following data (data likelihood components in Gadget terminology), from surveys and landing statistics, have been used in likelihood components within the estimation procedures:

- Summer trawl survey length distributions (MEDITS survey: Bertrand et al., 2003), aggregated to 1 cm length groups, from 1994 to 2007, were used in likelihood component LD1.
- Autumn trawl survey length distributions (GRUND survey: Relini, 1998) in 1 cm length groups, from 1994 to 2005 (1999 missing), were used in likelihood component LD2.
- Abundance indices from summer survey data (MEDITS survey: Bertrand et al., 2003), based on 3 length classes, from 1994 to 2007, were used in likelihood components SI1.1, SI1.2 and SI1.3.
- Abundance indices on autumn survey data (GRUND survey: Relini, 1998) based on 3 length classes, from 1994 to 2005 (1999 missing), were used in likelihood components SI2.1, SI2.2 and SI2.3.
- Official landings (IREPA), disaggregated by quarter for trawlers and mixed gillnets separately, from 2004 to 2007, were used in likelihood components LN.trw and LN.gill.
- Commercial catch length distributions (DCR: Data Collection Regulation) disaggregated for trawlers and mixed gillnets separately, in 2 cm and 4 cm length groups respectively, from 1998 to 2007 (with several gaps), were used in likelihood components LD.trw and LD.gill.

Modelled length distributions for both surveys and commercial fishery components (LD1, LD2, LD.trw and LD.gill) were compared to the respective likelihood data through the sum of squares (negative log-likelihood function):

$$l = \sum_{\text{time}} \sum_{\text{age}} \sum_{\text{length}} (P_{tal} - \pi_{tal})^2,$$

where P_{tal} is the proportion of the data sample for time/age/length combination t, a, l and π_{tal} is the corresponding proportion in the model population.

Three indices of abundance for each of the two experimental trawl surveys have been calculated. The length intervals have been considered from the observation of the length distributions from the surveys, in accordance with the first cohorts shown by the length structure (summer survey: SI1.1 = 2–12 cm, SI1.2 = 12–20 cm, SI1.3 = 20–80 cm; autumn survey: SI2.1 = 5–16 cm, SI2.2 = 16–26 cm, SI2.3 = 26–80 cm).

The likelihood components for the indices are the sum of squares of a log linear regression fitted to the difference between the modelled data (N_t) and the observed survey index (I_t). The slope has been fixed ($\beta = 1$), leaving Gadget to estimate the intercept.

$$l = \sum_{\text{time}} (I_t - (\alpha + \beta \ln N_t))^2$$

Landings data are compared with fleet disaggregated overall catches with a log sum of squares function. Data are disaggregated by quarter and cover the 2004–2007 time interval.

$$l = \sum_{\text{time}} \sum_{\text{fleet}} (\ln(N_{tf} + \varepsilon) - \ln(\nu_{tf} + \varepsilon))^2,$$

where N_{tf} is the catch data for that time/fleet combination and ν_{tf} is the modelled catch data for that time/fleet combination.

Model performance was also evaluated by comparing the basic conceptual model (mod) to alternative hypotheses and formulations such as:

- 1 recruitment in the first quarter (Alt1).
- 1 recruitment in the second quarter (Alt2).
- Estimating growth rate k under alternative assumptions of fishing effort (Alt3 and Alt4).
- Slow growth rate (Alt5).

2.2. Weighting procedure and parameters estimation

Regardless of the criterion adopted to fit model to the data, whether maximum likelihood methods, sum of squared residuals or Bayesian methods, when multiple sources of information are combined into an overall sum of squared errors, each component will contribute proportionately to its associated relative weight. Several weighting protocols have been developed, but most of them generally try to consider the degree of uncertainty associated with the data or the statistic used (Haddon, 2001).

In what follows, the weighting method proposed by Stefansson (1998, 2003) is used where the minimum sum of squared error (sse) for each component is obtained through the estimation procedure applied to a very flexible model with each component one by one heavily weighted. The flexibility of the model is obtained by permitting Gadget to estimate most of the parameters (see Stefansson (2003) and Taylor et al. (2007) for more details on the iterative reweighting scheme). The inverse of these minimum sums of squared errors, multiplied by the degrees of freedom for each component, is used as a variance estimate to build the relative weight for each component. This implies that the greater the variability, the less the relative weight. The degrees of freedom represent the number of terms that constitute the data and are the number of potentially informative length cells for each year/step/age cell.

In the development of large models, it is generally a matter of choice to decide what should be estimated within the model and what should be taken as an assumption or externally estimated. This is also what has been done here for the hake model in the Gadget framework. One of the main reasons for not leaving parameter estimation completely to Gadget for the large number of biological processes, was to reduce the number of parameters to deal with. This was achieved through assumptions that constrained the model on some aspects of minor interest for the current work or for which more reliable information were available. For this reason estimation was mostly restricted to the scale factor for the initial population, the scale factor to size recruitments, fishing effort and growth rate.

2.3. Model projections and alternative management measures testing

Deterministic projection models produce only a single realisation or projection of the development of the system. Nevertheless, some sources of stochasticity can be taken into account in these forward simulations. Multiple sources of uncertainty have been recognised in modelling biological systems and processes (Charles, 1998; Harwood and Stokes, 2003) and simulations invariably include stochastic elements in which random variation around one or more model components can be included.

Table 1
Component weights: the inverse minimum sums of squares, the scaled weights calculated from the iterative reweighting procedure and the ratio of the sum of squares from the optimised model (ssef) with the minimum from the weighting run (ssem).

comp	1/ssem	ssef/ssem	scaled *Wi*
LN.trw	2.682	31.939	43
LN.gill	0.640	1.992	10
LD.trw	1.065	1.932	486
LD.gill	1.744	1.997	436
LD1	4.405	1.155	2344
LD2	6.378	1.654	2666
SI1.1	36.928	22.548	517
SI1.2	4.405	34.392	62
SI1.3	0.808	2.949	9
SI2.1	0.267	1.812	4
SI2.2	1.289	9.902	14
SI2.3	7.205	8.242	79

A Monte-Carlo approach was used to simulate forward projections of the model taking into account the uncertainty related to recruitment and evaluating its implications on the model under different management scenarios. Resampling was carried out assuming recruitment as gamma distributed (Shelton, 1992; Taillie et al., 1995) according to the recruitment variation observed in the 1994–2007 time interval, which is in accordance with statistical tests of these assumptions (Kolmogorov–Smirnov: $D = 0.160$, $p = 0.868$). Running the forward simulations a sufficient number of times (limited to 100 simulations here for computational reasons) summary information was obtained in the form of confidence intervals associated with the predictions, thus an estimation of uncertainty. Several such 8-year forward projections of the modelled hake stock were done assuming a stable fishing effort based on 2007 levels.

The effect of closed area management scenarios, designed to reduce fishing mortality on recruits during their highly vulnerable phase of aggregation in nursery areas (Ardizzone and Corsi, 1997; Colloca et al., 2009), was evaluated on the basic model. These included simulation of two different scenarios for protecting of hake below 15 cm in total length (Bartolino et al., 2008b) from the trawl fishery (40% and 60% levels) and the effects on the expected adult stock biomass (SSB) were evaluated.

3. Results

3.1. Basic hake model

The initial inverse weights and the weights determined from the iterative procedure are given in Table 1 along with the ratio between the sum of squared errors (sse) from the final model and the minimum for each component. When combining information from all the different data sources, the final sse for each component is of course greater than the minimum during the iterative reweighting.

LD1 and LD2 likelihood components have the highest weights as expected from experimental data. Indices of abundance from the two surveys reveal an opposite weighting according to their fitting to the model dynamics, summer survey indices high weight the small hake length group SI1.1 while autumn survey indices high weight the large hake group SI2.3.

The estimated length distributions for both the surveys are dominated by the first cohort that produces an almost symmetric dome shape curve. During low recruitment years a second cohort appears in the summer survey component. As expected the first cohort mode is shifted from 8–9 cm to 11–12 cm from the summer to the autumn survey. This growth pattern appears correctly represented by our basic model. Residuals for length distributions presented homogeneous dispersion for both trawl surveys for small hake. The model tends to over-predict mid-length and large fish.

Table 2
Estimated fishing mortality with 95% confidence interval by age class.

Age	$F \pm 95\%$ CI
0	0.70 ± 0.19
1	1.02 ± 0.28
2	0.31 ± 0.10
3	0.29 ± 0.10
4+	0.29 ± 0.10

The model has a very high fit (low sum of squared error) for SI1.1 likelihood component (Fig. 4) as expected from the high weight associated to this component (Table 1). The indices from the two surveys showed comparable sum of square errors, thus fitting, for the intermediate length class, while the large hake group found better agreement with the autumn survey index SI2.3 (Fig. 4). The intercept of the assumed linear relationship can be interpreted as a relative measure to compare catchability between length groups and surveys. Length groups 2 and 3 show lower intercepts respect length group 1, as expected according to lower catchability of large fish. No sensible differences were found in the catchability of the two surveys.

Estimated length distributions from the commercial fleet components were found in agreement with data for most of the time steps available (Fig. 5). The model simulates two distinct cohorts of trawl landings in time step 2 as generally observed in the data, with the second always larger than the first although large variations are observed in the data about their respective size. Length distributions of gillnet landings show a relatively simpler structure with a main mode at approximately 30 cm in fish total length.

Recruitment was spread between time steps 1 and 2 with an estimated comparable size (only 5% difference, Fig. 3). Estimated fishing mortality F showed highest values in the first two age classes, as expected for a fishery mostly targeting immature fish, with a peak value of 1.02 ± 0.28 for age 1 hake (Table 2).

Spawning stock biomass (Fig. 6) showed a strong decrease during the first five years (1994–1998) of the period investigated. We found a moderate reduction in biomass in the following period and a minimum in 2003 with an estimated value less than half of the initial value of 1994.

3.2. Model comparison

Basic model (mod) overall likelihood score was the second best value after Alt5, with a value of 4284 against 4126. The other formulations range between 4448 and 7858. Models were compared and evaluated according to their fit to the data but also considering the differences in the estimated population dynamics (Fig. 6). Model runs can be divided in two distinct groups according to the estimated spawning stock biomass. Mod, Alt1 and Alt2 estimated much higher values of SSB than the Alt3, Alt4 and Alt5 formulations, but at the same time lower relative reduction of mature fish throughout the time period investigated, approximately 50% reduction against 70–75% reduction. Due to the single recruitment event assumption both Alt1 and Alt2 required the estimation of large number of recruits coming into the models in a single time step to ensure the survival of a minimum number of fish to fit landings in the third and fourth quarter of each year. This penalized fitting of length structure components LDs from both fisheries and surveys (Fig. 7).

Those model alternatives including growth rate parameter optimisation (Alt3, Alt4, and Alt5) had some improvements in fitting the survey-related likelihood components but they totally failed in

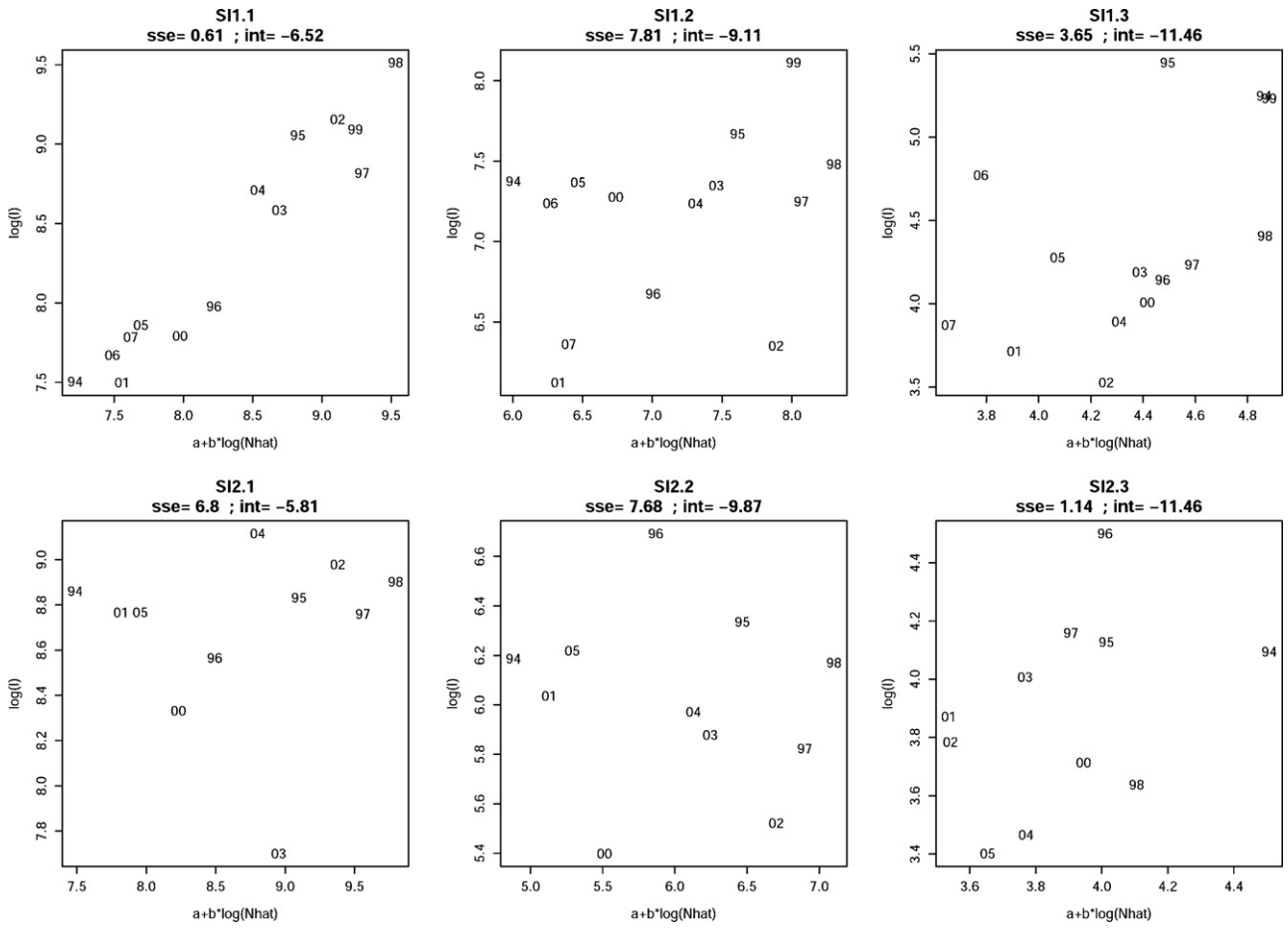


Fig. 4. Plots of the log linear regression between survey indices from likelihood data and as estimated by gadget. $\ln(I)$ is the survey index and $\alpha + \beta \ln N_t$ is the fitted regression to the modelled fish population.

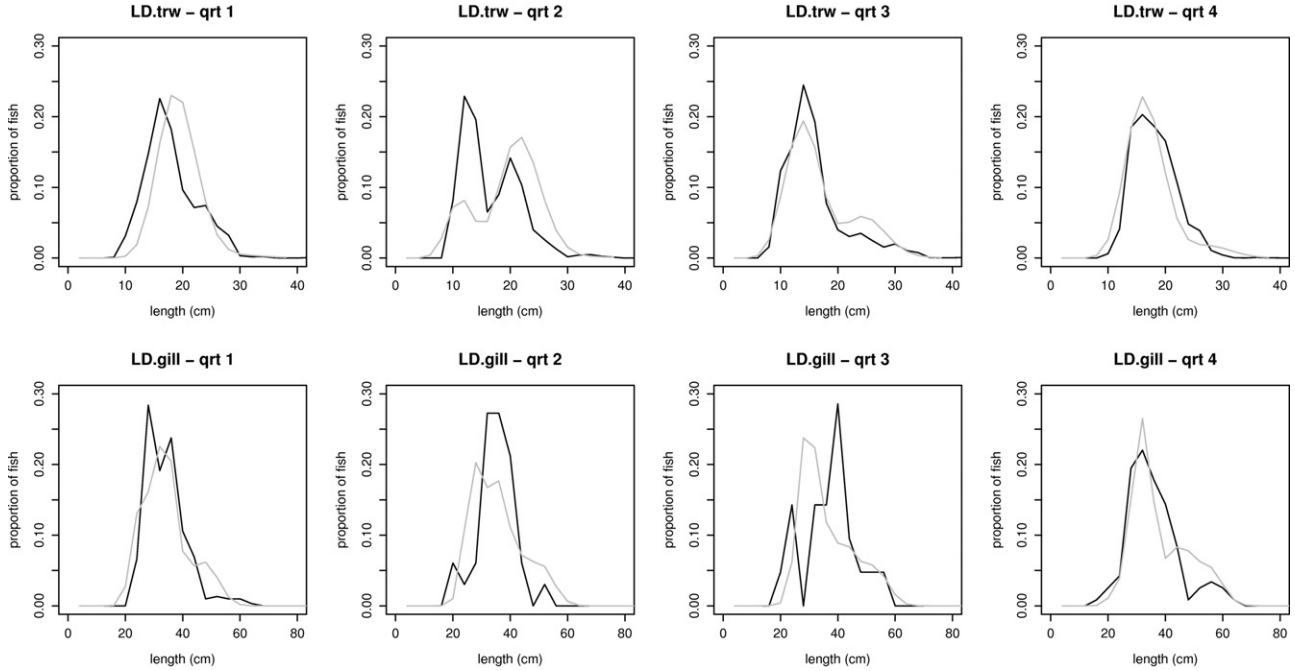


Fig. 5. length distribution of four time steps for 1998 trawler landings (first line) and 2006 gillnets (second line) from data (black line) and gadget basic model (gray line).

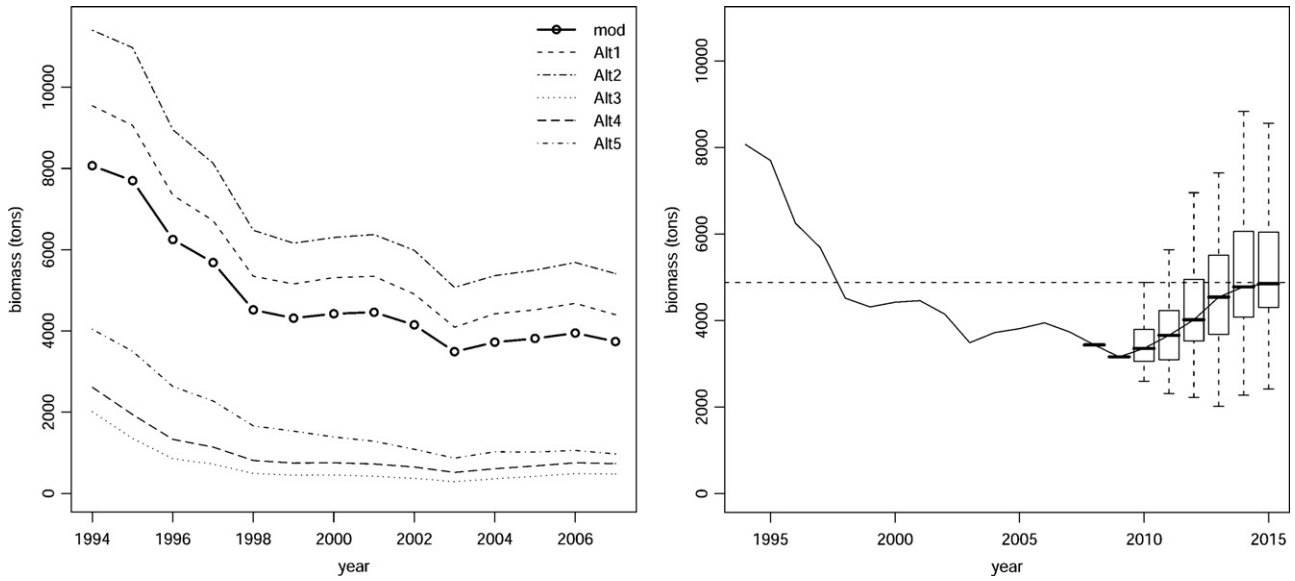


Fig. 6. (a) Estimated spawning stock biomass (timestep 4) from the basic and alternative models from 1994 to 2007. (b) Annual estimated spawning stock biomass (timestep 4) with boxplot of projections up to 2015 under no management scenario, dotted horizontal line mean estimated SSB between 1994 and 2007.

fitting commercial catches, underestimating appropriate landings of large hake by gillnet fishery.

3.3. Model projections and nurseries closed areas scenarios

8-Years forward projections of the modelled hake stock have been done (Fig. 6), assuming a constant fishing effort based on 2007 data. After a further decrease in 2008 and 2009 with a minimum of 3161 t, SSB was estimated to have a progressive increase in the following years to reach 4853 t in 2015. As expected forward estimations were characterized by consistent variability as consequence of the high uncertainty associated to recruitment predictions (Fig. 6). The degree of uncertainty associated to the SSB estimations increased particularly in 2012 and 2014. Reduction of 40% and 60% of only recruits fishing mortality produced moderate effects on the predicted SSB from 2010. In 2015, after 8 years of simulated management scenarios, SSB was found 3–5% higher than in the scenario with no additional protection on recruits.

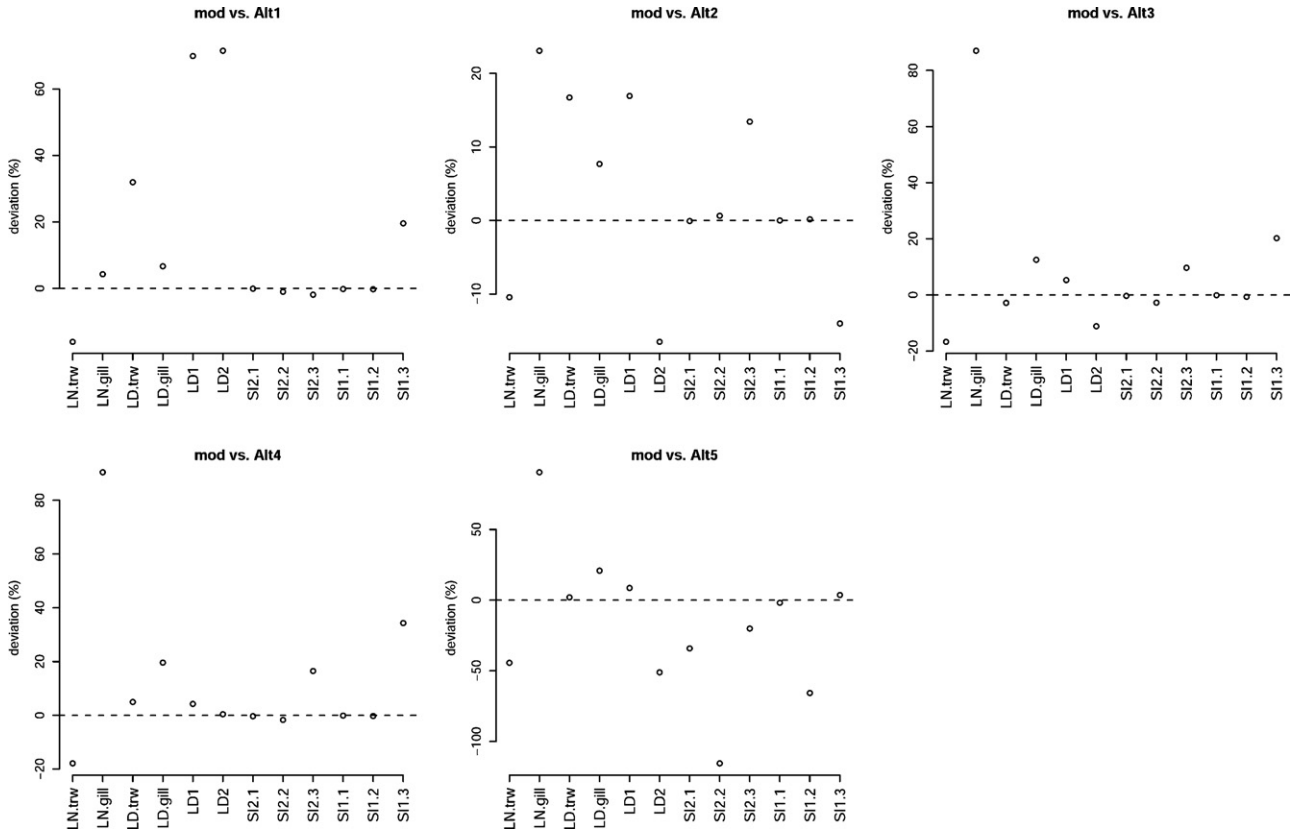


Fig. 7. Likelihood score deviations by component of the alternatives from the basic model.

4. Discussion

The ability of Gadget to model several sub-processes and mechanisms, combined with its flexibility, offered an extensive control of a large number of model components. The modelling framework performed successfully also in a commercial landing data limited context, common for the Mediterranean. Weighting the different informations available gave the possibility to prioritise the importance of the various data sources, highly weighting in an objective manner specifically designed experimental surveys and lowly weighting noisy commercial fisheries data. This framework allowed considering also the available official statistics of commercial landings, known for being highly problematic due to the atomised characteristics of the Italian fisheries (SGMED, 2008) and for this reason rarely included in modelling approaches. Although their associated low weight, commercial landing components played a fundamental role in model selection because complemented the lack of information on the adult portion of the stock that was poorly represented by the only scientific trawl surveys.

Models with a single recruitment per year obtained worse scores than those modelling two annual recruitment events supporting results from previous studies that described hake as a protracted spawning species (i.e. Maynou et al., 2003; Goni et al., 2004).

As expected the use of fast growth parameters for hake, almost twice respect those generally applied in previous assessments (i.e. SAC, 2003) had a strong impact on the modelled dynamics and represented an unprecedented improvement of modelling hake population structure in the Mediterranean. Assumptions and processes more congruent with current knowledge on the biology of the species should be considered the first step toward more realistic reconstructions of the recent history of stocks and to evaluate their current status.

The reduction in the fishing effort observed during the last years (also reported in SAC, 2003), coupled with fast recovery features of the hake stock (*r*-selection), have the potentiality for a moderate increase of the stock in the next years. Such possible increment in the stock size should be interpreted correctly at the light of the available short time series of data. Considering that high exploitation of this stock started long time ago, during '50s in the post-war period (Matta, 1954) and that signs of overfishing were detected since '80s (SGMED, 2008), the increase in the population biomass predicted by our simulation is likely to be far from natural pristine conditions anterior to fisheries exploitation.

The current high levels of fishing mortality on hake juveniles, with an F value of approximately 1.02 for age 1 fish, is unlikely to be compatible with a responsible, sustainable and optimised exploitation of the hake stock as also reported by previous assessments (SAC, 2003; SGMED, 2008). Moreover low recruitment levels have been observed after warm winters in association with low primary productivity springs and in years characterized by particularly high temperatures in summer (Bartolino et al., 2008a). This suggested an increasing risk of consecutive low annual recruitments under a global warming scenario that would reduce the recovery abilities of the stock observed during the last few years and assumed for projecting our model.

Although the high variability of hake recruitment and the particularly high fishing mortality that characterized the first age classes, our model suggested that a reduction in F only for recruits would have moderate effects on the stock biomass trajectories. Colloca et al. (2009) demonstrated the occurrence of extremely high concentrations of hake recruits in the Tyrrhenian and Ligurian Sea (60% of recruits are concentrated in only 10–11% of the area). The elevated spatio-temporal stability of the hake nurseries in this area (Colloca et al., 2009) would guarantee the effectiveness throughout time of the implementation of closed areas. In spite of this, the present population structure model showed only minor effects on SSB even in the most precautionary management scenario (protection of 60% of the recruits). Management measures specifically designed to protect only hake recruits would have only a limited impact in terms of fish that would be able to grow up to the size of maturity. This can be explained by the fact that beside the elevated fishing pressure on recruits, the total mortality of age-0 hake is still dominated by the high natural mortality. In any case, the interpretation of the outcomes should take into account the various limitations of the model presented. For instance, the current model implementation lacks a relationship linking the number of recruits to the parental stock and to the environmental variability. Moreover, additional sources of uncertainty related to our knowledge of key processes such as natural mortality or fisheries selectivity can have important effects on our ability to project stock trajectories for the next future.

The results in this paper should not be taken as a final and definitive conclusion that closures of areas to trawling will have insignificant effects on the stock. This whole topic is currently under considerable debate within the scientific community. On the one hand there is a fairly common perception that closed areas can be an effective tool for marine conservation and management (e.g. Lubchenco et al., 2003) but on the other hand there are several analyses and simulations which indicate the exact opposite (see e.g. Stefansson and Rosenberg, 2005, 2006 and references therein). For this reason it is essential that proposals for closed areas are evaluated along the lines recommended in Hilborn et al. (2004). These evaluations need to be as stringent as evaluations of other management measures such as quota controls etc, which have a long history of being evaluated quantitatively, in contrast with many suggested area closures which have often not been based on taking the population dynamics or uncertainty carefully into account. The present paper and that of Colloca et al. (2009) should therefore be considered only the first attempts at such evaluations. Results here imply that the positive effects which might be expected from protecting hake nursery grounds are only marginally linked to a reduction in hake recruits fishing mortality, which follows from taking into account the high natural mortality on these juveniles, a perfectly reasonable result. On the other hand, the 5% increase in the SSB projections under the spatial closure scenario may well be an underestimate of the actual effect that closed areas would have for hake stocks in the study area. For example, other age classes of hake, which occur in these same areas of recruit aggregation, would benefit by a reduction of fishing activity on such grounds. Moreover, trawl fisheries have been shown to have severe impact on benthic fauna where commercial fish aggregate (Koslow et al., 2000). A decrease in trawling impact over hake nurseries would ensure preservation of associated habitats of the shelf-break such as crinoids seabed (Colloca et al., 2004) that have been shown to play a key role in structuring important highly productive demersal assemblages in the Mediterranean (Cartes and Carrassón, 2004; Cartes et al., 2007; Colloca et al., 2003). At the current stage the effect of high quality shelf-break habitats on hake recruits survival is almost unknown and our hake model does not include species interaction, but the high impact of trawling on benthic communities may well have a negative impact on associated demersal species such as hake through the food-web. Such results would invalidate the effects given in Stefansson and Rosenberg (2005) since they correspond to increasing r in an r–K model and could potentially give much higher impact than previously believed.

It has been shown here how the development of population structure models including multiple processes is possible also in a data-limited context, when diversified information on the biology and life history of species is available such as for hake in the Mediterranean. The fragmentation of the information available rep-

resents a challenge for modelling population dynamics and possibly a limit to the application of a complex modelling framework. However, as presented here it did not preclude the development of a model that contributed to improve understanding of the hake stock dynamics and that suggested interested unexpected effects of recruit protection. The model included several key processes which regulate the hake stock dynamics but also relies on assumptions and (over-)simplifications of the real world. Collection of new data and research effort can contribute to refine and improve the basic model presented here, to more closely respond to the specific needs of management.

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